Jaiveer Bagga

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Education

Georgia Institute of Technology Atlanta, GA

B.S. Computer Engineering; 4.0 GPA *Expected Graduation May 2028*

- Relevant Coursework: Data Structures and Algorithms, Linear Algebra, Discrete Math, Differential Equations, Multivariable Calculus, Digital System Design, Mechanics, Electricity & Magnetism

Technical Skills

Languages: Java, Python, TypeScript, C/C++, SQL
Developer Tools: Git/GitHub, Docker, PostgreSQL, AWS, Arduino IDE, Raspberry Pi
Frameworks: React, FastAPI

Work Experience

X Cap Market Jun 2025 – Aug 2025

Software Engineering Intern

- Built a web scraping pipeline in Python using the OpenAI API, profiling 2,000+ private equity and investment firms for lead generation into DynamoDB, with retry logic and rate-limit handling
- Implemented the frontend of a full-stack client management platform using React and TypeScript
- Designed a single interface consolidating customer records and outreach emails for XCap's clients
- Developed backend APIs in Python, deployed via AWS Lambda and containerized with Docker

Projects

Portable Air-Quality Filter/Monitor Jul 2025

University of Pennsylvania M&TSI | Arduino, 3D Printing, TypeScript, Xcode

- Led engineering within a five-person interdisciplinary team for a portable air quality detection/filtration device with an airflow of 22.7 ft^3/min, integrating 3D-printed housing and Arduino-based sensors
- Built an app using TypeScript and Xcode displaying live sensor readings from the device
- Formulated go-to-market plan projecting 500 customers, $132K Year 1 revenue, and 7:1 CLV:CAC

Eye Disease Detection App Oct 2024 – Dec 2025

Developer | Machine Learning, Python, App Development

- Built an ML app to predict cataracts, glaucoma, and diabetic retinopathy, achieving 96% accuracy
- Tested with 15 patients alongside an ophthalmologist; won 3rd place, Congressional App Challenge

Fall Recognition App Aug 2024 – Apr 2026

Developer | Raspberry Pi, Machine Learning, Python, Computer Vision

- Developed Steadymatch, a real-time fall detection system using an 11.2M-parameter YOLOv8 model
- Engineered pose-based heuristics using torso angle, bounding-box geometry, and normalized body position
- Classified falls from normal movement and triggered instant caregiver alerts

Research

Predicting Maximal Oxygen Uptake Using LSTM Models and Physiological Signals October 2025

Sole Author

- Designed and trained a two-layer LSTM model in PyTorch to predict VO_2max from physiological time-series data
- Achieved an R^2 of 0.9693 and RMSE of 0.1961 mL/min on the test set
- Outperformed a linear regression baseline, which achieved an R^2 of 0.8621
- Engineered a 20-second rolling-average target variable (AO20) and optimized hyperparameters via random search with 5-fold cross-validation